

Om Patel

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Education

Dhirubhai Ambani University (DAU), Gandhinagar

2026 – 2028

M.Tech in Information and Communication Technology (Machine Learning)

Institute of Infrastructure Technology Research and Management (IITRAM), Ahmedabad

2022 – 2026

B.Tech in Computer Engineering

CGPA: 9.64/10

Chairman & Gold Medalist

Internships

Dhirubhai Ambani University (formerly DA-IICT), Gandhinagar

May 2025 – July 2025

Summer Research Intern

Certificate [🔗](#)

- Analyzed **two-wheeler driving patterns** using smartphone sensor data for rider behavior and road anomaly detection.
- Synchronized multi-sensor data into an **orientation-independent** format and applied DBSCAN, Wavelet Filtering, and DTW.
- Built a behavior analysis pipeline using **Fuzzy Logic, Random Forest, and HMM**, with interactive map visualization.

YADS Technologies Pvt. Ltd.

June 2024 – July 2024

Python Full Stack Developer Intern

Certificate [🔗](#)

- Integrated multiple APIs to collect and merge keyword-based public, news, and social media data into PostgreSQL.
 - Designed SQL queries for efficient retrieval and processing of collected datasets.
 - Applied **sentiment analysis using Python** to analyze collected textual data.
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Projects

Trinetra: Edge-AI Wearable Assistive System

GitHub [🔗](#)

- Technologies:** Flutter, Python, Flask, FastAPI, YOLOv11, Qwen2-VL, Raspberry Pi 5, WebSocket, OCR, GPS, Ultrasonic Sensors
- Developed an **Edge-AI and IoT wearable system** for visually impaired users using object detection, sensor fusion, voice interaction, and guardian-assisted emergency support.
- Implemented **INT8 YOLOv11** inference, local OCR, ultrasonic obstacle detection, GPS, and remote servo-controlled camera assistance.
- Research paper submitted to IEEE Transactions on Human-Machine Systems (THMS).*

Course Management System

Live App [🔗](#) — GitHub [🔗](#)

- Technologies:** React, Node.js, Express, MongoDB
- Built a full-stack academic platform with **Admin, Faculty, and Student** modules for attendance, quizzes, results, fees, assignments, and forums.
- Implemented **role-based authentication, REST APIs, database integration**, and administrative report generation.

ChitChat – Real-Time Chat Application

Live App [🔗](#) — GitHub [🔗](#)

- Technologies:** React, Node.js, Express, MongoDB, Zustand, Socket.IO
- Developed a real-time chat application supporting **direct/group messaging, file sharing, authentication**, and emoji support.
- Implemented **Socket.IO-based communication** and integrated frontend, backend, and database services.

ClipboardSync – Cross-Platform Clipboard App

GitHub [🔗](#)

- Technologies:** React Native, Electron.js, Node.js, Socket.IO
 - Built a cross-platform application to sync **text and image clipboard content** between mobile and desktop devices over Wi-Fi.
 - Implemented WebSocket communication, clipboard history, and local in-memory storage without storing clipboard data on remote servers.
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Technical Skills

- Programming Languages:** C/C++, Python, JavaScript
 - Machine Learning & Data Science:** TensorFlow, Keras, Pandas, NumPy, Scikit-Learn, NLTK
 - Web Development:** ReactJS, NodeJS, Express.js, HTML, CSS, Tailwind, Bootstrap, REST APIs
 - Databases:** MongoDB, PostgreSQL, MySQL
 - App & Systems:** React Native, ElectronJS, Flutter, FastAPI, Flask, WebSocket, Socket.IO, Git
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Achievements

- 4th Place, Road Safety Ideathon 2025**, PDEU, Gandhinagar, among 25 finalist teams.
- Chairman & Gold Medalist**, B.Tech Computer Engineering, IITRAM.
- Participated in **Flipkart GRiD 6.0, HACKOUT 2025**, and **IIT Gandhinagar x Odoo Amalthea 2025**.